MUHAMMED BASITH N

Mechanical Engineer(Embedded System & ML developer)

Kozhikode, Kerala, India

Professional Summary

Mechanical Engineer transitioning into Embedded Systems, Automation, and Machine Learning. Combines a deep academic foundation in physical mechanics, fluid dynamics, and thermal systems with hands-on experience in IoT hardware prototyping, Python-based AI microservices, and PCB assembly. Adept at bridging the gap between mechanical hardware and intelligent software logic to drive positive technical outcomes.

Education

Bachelor of Technology (B.Tech) in Mechanical Engineering
2022 – 2026 ***TKM College of Engineering*** - Kollam, Kerala

Experience

R&D and Operation Aug 2024 - Apr 2026
SpinZone, TKM College of Engineering

- ✓ Collaborated directly with software engineers to map out system architecture, design embedded automation logic, and debug runtime code for smart laundry machines.
- ✓ Worked alongside hardware designers to assemble electronics, solder micro-controllers and custom hardware boards, and identify hardware-level faults.
- ✓ Managed physical deployments, executing the electrical installation and network configuration of automated washing machines across multiple hostel locations.

Featured Technical Projects

AI Fraud Detection Pipeline (MVP)
Python, FastAPI, Scikit-learn, Cloud

- ✓ Architected an end-to-end Machine Learning microservice that simulates user behavior, trains a logic model, and serves predictions.
- ✓ Programmed a Data Simulator to bypass PII restrictions, intentionally injecting imbalanced fraud patterns (99:1 ratio)
- ✓ Shifted evaluation metrics from pure accuracy to a Confusion Matrix (Precision/Recall) to accurately measure threat detection versus false alarms
- ✓ Deployed a production-ready REST API via Render, implementing joblib for model serialization and a SQLite ledger for audit logging

Cobweb Channel Cold Plate Analysis
ANSYS Fluent, CFD, Thermal Analysis

- ✓ Conducted advanced numerical analysis of a cobweb channel design using ANSYS Fluent CFD software.
- ✓ Optimized the thermal performance and heat dissipation capabilities of porous media cold plates

Personal Info

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Indian

Languages

- ✓ Malayalam, native
- ✓ English, fluent

Skills

- ✓ Embedded Systems & Automation
- ✓ Machine Learning Pipelines
- ✓ PCB Soldering & Assembly
- ✓ Hardware Configuration & Troubleshooting
- ✓ CAD
- ✓ Mechanics
- ✓ Thermal Analysis

Embedded & Software

- ✓ Python
- ✓ ESP32
- ✓ C Programming
- ✓ Scikit-learn
- ✓ FastAPI
- ✓ Cloud Deploy
- ✓ ANSYS Fluent

Extracurricular

- ✓ APJ KTU Interzone Chess Champion

Scan for Portfolio